October 20, 2026

Dear Editors of *Algorithmica*,

I submit the manuscript *"Cluster Editing above a P_3 packing: W[1]-hardness in the excess and the treewidth of the intersection graph"* for consideration as a research article.

Summary. Li, Pilipczuk and Sorge parameterize Cluster Editing by the excess ℓ of the solution size over the size of a modification-disjoint packing H of induced P_3's, and ask for which packings the problem is fixed-parameter tractable. The manuscript uses the treewidth w of the intersection graph of the packing as the structural measure. The main result is negative: the problem is W[1]-hard parameterized by $\ell + w$, by $\ell +$ pathwidth and by $\ell + c$ (where c is the maximum number of packed P_3's through a vertex), even for pair-maximal packings. Under the Exponential Time Hypothesis there is no algorithm with running time $f(\ell, w)\, n^{o(\sqrt{\ell})}$. The reduction is from Grid Tiling with $\leq$. On the positive side, a constraint-satisfaction formulation gives algorithms with running time $3^{w+1}n^{O(1)}$ for $\ell = 0$ and $5^{w+1}n^{2\ell+O(1)}$ in general, and the manuscript identifies a configuration, the soft triangle, whose absence makes the case $\ell = 0$ polynomial for every c and w. Further structural results and open questions are stated.

Relation to earlier work. The problem and the parameter ℓ are due to Li, Pilipczuk and Sorge, and their question is the starting point of this work. For vertex-disjoint packings, fixed-parameter tractability in ℓ is due to van Bevern, Froese and Komusiewicz; I use their reduction rule on the components of the intersection graph and cite its soundness proof. Blaha et al. (arXiv:2602.15683) show that Cluster Editing is fixed-parameter tractable in the treewidth of the input graph; this is a different parameter from the treewidth of the intersection graph used here. No part of the manuscript has been published before, and it is not under consideration elsewhere.

Verification. The hardness reduction is computer-assisted at four finite steps, which the paper lists. Its counting core and several structural lemmas are machine-checked in Lean 4; the Lean files do not cover the graph-level statements of the reduction. Code and Lean files are available at `https://github.com/vinhqdang/correlation_clustering` (directories `paper2_p3_interaction/code` and `paper2_p3_interaction/lean/P3Proofs`).

Length. The manuscript is long (113 pages in the journal template). The main text states every result, describes the hardness reduction and the algorithms, and gives the key proofs; the long proofs of the reduction lemmas, the analysis of soft triangles, and further structural results are in the appendices. I am glad to shorten or reorganize the appendices, or to move them to an online resource, if the editors prefer.

I have no competing interests to declare.

Sincerely,

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